To find a factor (nontrivial) of n

Pick an Elliptic curve

$$E = \{ (x,y) \in \mathbb{Z}_n^2 \mid y^2 = x^3 + Ax + B \} \cup \{\infty\}$$

[We have to pick A & B].

Choose $(a,b) \in \mathbb{Z}_n^2$ & $A \in \mathbb{Z}_n$ & find

$$\underline{B = b^2 - a^3 - Aa}$$

Algorithm (Lenstra)

1. Choose $A, a, b \pmod{n}$
2. Set $P = (a,b)$ & $B = b^2 - a^3 - Aa \pmod{n}$
3. Loop $j = 2, 3, \dots$ up to a specified bound.
4. $Q = jP \pmod{n}$ & set $P = Q$
5. If the computation in step 4 fails, then you get proper factor d of n.
6. If $d < n$ & return d.
7. Else $d = n$, then go to 1 & choose a new set $\{A, a, b\}$
8. Else Go to 3.

For $n = \underline{589}$, $a=2, b=5$, $A=4$

Then $B = b^2 - a^3 - Aa = 9 \pmod{589}$

So $E = \{(x,y) \in \mathbb{Z}_{589}^2 \mid y^2 = x^3 + 4x + 9\} \cup \{\infty\}$

$2P = (564, 152)$

$3!P = 3(2P) =$ ~~$(P+2P)$~~

$4!P = 4(3!P) = (396, 209)$

$5!P = (\quad , \quad)$

for this $m = \frac{3x^2+4}{2y} = \frac{3\times 396^2+4}{2\times 209} \pmod{589}$

$= \frac{3\times 142+4}{418}$

$589 = 418\times 1 + 171$

$418 = 171\times 2 + 76$

$171 = 76\times 2 + 19$

$76 = 19\times 4 + 0$

So $GCD(418, 589) = 19$

& hence $\underline{19 \mid 589}$

- Find k s.t. $k!P$ is not defined

if no such k or for such a k divisor is not proper choose a different Elliptic curve.

Recall that to a prime divisor of n (or equal) we need to consider all primes less than $\sqrt{n}$.

(Primality test)

Proposition: Let $n (>1) \in \mathbb{N}$. Let q be a prime s.t. $q > \sqrt{n} - 1$ & $q \mid (n-1)$. If there exists an integer a s.t.

1. $a^{n-1} \equiv 1 \pmod{n}$
2. $\gcd(a^{\frac{n-1}{q}} - 1, n) = 1$ ✓

Then n is a prime.

Proof: Suppose to contrary n is composite & p is a prime that divides n so that $p \leq \sqrt{n}$.

$q > \sqrt{n} - 1 \geq p-1$ so that $q > p-1$ & hence $\gcd(q, p-1) = 1$.

Let $u, v \in \mathbb{Z}$ s.t. $\underbrace{uq + v(p-1)} = 1$. Then

$$a^{\frac{n-1}{q}} = a^{\{uq + v(p-1)\}\left(\frac{n-1}{q}\right)}$$

$$= a^{u(n-1)} \times a^{v(p-1)\left(\frac{n-1}{q}\right)}$$

$$= \left(a^{n-1}\right)^u \times \underbrace{\left(a^{p-1}\right)}^{v\left(\frac{n-1}{q}\right)}$$

$$= \underbrace{1} \times 1 \pmod p$$

$$a^{\frac{n-1}{q}} = 1 \pmod p$$

$a^{n-1} \equiv 1 \bmod n$

$a^{n-1} = a$

$n = pk$

$n-1 = pk-1$

$\boxed{p \mid n \;\&\; a^{n-1} \equiv 1 \bmod n}$

$\Rightarrow a^{n-1} \equiv 1 \bmod p$

$p \mid \gcd\left(a^{\frac{n-1}{q}} - 1, n\right)$ ~ contradicts ~~[illegible]~~ cond^n (2).

$a \equiv a^n = a^{kp} = (a^p)^k = a^k$

$n = kp$

$a^{n-1} = 1 + \cancel{k}\, l n$

$= 1 + lkp$

$= 1 \bmod p$

This is called Pocklington-Lehman test

Propn: Let $n \in \mathbb{N}$ & let

$$E = \{(x, y) \in \mathbb{Z}_n^2 \mid y^2 = x^3 + Ax + B\} \cup \{\infty\}$$

Let $m \in \mathbb{N}_{\times}$ let q be a prime s.t. $q \mid m$ & $q > (n^{1/4} + 1)^2$. If $\exists$ a pt $P \in E$ s.t.

1. $mP = \infty$ ✓

2. $\left(\frac{m}{q}\right)P \neq \infty$

is defined, then n is a prime.

$$\boxed{\text{Hasse's bound } \quad \left|\#\left(E(\mathbb{F}_q)\right) - (q+1)\right| \leq 2\sqrt{q}}$$

In particular when $q = p$ a prime.

Proof: Suppose n is composite & let p be a prime divisor.

~~By Hasse's bound~~, if $\#\left(E(\mathbb{Z}_p)\right) = m_p$

By Hasse's bound $\quad |m_p - (p+1)| \leq 2\sqrt{p}$

or $\quad m_p \leq p+1+2\sqrt{p} = (\sqrt{p}+1)^2 \leq (n^{1/4}+1)^2 < q$

So $\quad \underline{1 \leq m_p < q}$

So $\quad \gcd(m_p, q) = 1 \quad$ (since q is a prime)

$1 = uq + v\, m_p$

$\Rightarrow \quad \underline{1 = uq \mod m_p}.$

| $E(\mathbb{Z}_n) = E_n$ | $E(\mathbb{Z}_p) = E_p$ |

For $P = (x, y) \in E_n = E$, you get $Q = (x', y') \in E_p$

$x \equiv x' \mod p$

$y \equiv y' \mod p.$

| Eg. $5 \in \mathbb{Z}_6 \rightsquigarrow 2 \in \mathbb{Z}_3$

$$q \mid m \Rightarrow \frac{m}{q} \in \mathbb{Z} \Rightarrow \left(\frac{m}{q}\right)Q = uq\left(\frac{m}{q}\right)Q = (um)Q = u(mQ)$$
$$= u\infty$$
$$= \infty$$

$$\boxed{P = \infty \iff Q = \infty}$$ ✓

$\rightarrow \cancel{Q'} \left(\frac{m}{q}\right) Q = \infty$ is a contradiction

because $\underbrace{\left(\frac{m}{q}\right) P}_{= P'} \neq \infty$
